

ML API Deployment Project

Project Description

This project is a simple Machine Learning API created using Flask and Docker.

The API uses a trained Machine Learning model to predict the iris flower type based on flower measurements.

Technologies Used

- Python
 - Flask
 - Scikit-learn
 - Joblib
 - Docker
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Project Files

- model.py → Trains the Machine Learning model and saves it
 - model.pkl → Saved trained model file
 - app.py → Flask API application
 - Dockerfile → Docker setup configuration
 - requirements.txt → List of required Python libraries
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Prerequisites

Before running the project, make sure the following software is installed:

1. Python (optional if using Docker)
2. Docker Desktop

Download Docker Desktop from:

<https://www.docker.com/products/docker-desktop/>

After installation, open Docker Desktop and make sure it is running.

How to Run the Project Using Docker

Step 1 — Open Terminal

Open Command Prompt, PowerShell, or VS Code terminal.

Navigate to the project folder:

```
cd path_to_project_folder
```

Example:

```
cd Documents/ML-API
```

Step 2 — Build Docker Image

Run the following command inside the project folder:

```
docker build -t ml-api .
```

This command creates a Docker image named ml-api.

Wait until the build process completes successfully.

Step 3 — Run Docker Container

Run the following command:

```
docker run -p 5001:5000 ml-api
```

Explanation:

- 5001 → Port on your computer
- 5000 → Port used inside the Flask application

If successful, the Flask server will start running.

Testing the API

You can test the API using:

- Thunder Client (VS Code extension)
 - Postman
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API Endpoint

Prediction Endpoint

POST http://127.0.0.1:5001/predict

Sample Request Body

Send the following JSON data in the request body:

```
{  
  "features": [5.1, 3.5, 1.4, 0.2]  
}
```

Sample Response

If the request is successful, you will receive a response like this:

```
{  
  "prediction": 0  
}
```

Notes

- Docker Desktop must be running before executing Docker commands.
- Make sure you run commands inside the project folder.
- If port 5001 is already in use, change it to another port.

Example:

```
docker run -p 5002:5000 ml-api
```

Then use:

```
http://127.0.0.1:5002/predict
```

Notes

- Docker Desktop should be installed before running the project.
- Thunder Client or Postman can be used to test the API.